



United International University

B.Sc. in Computer Science and Engineering (CSE)

CSE 4891: Data Mining

Midterm Exam : Summer 2025 Time: 1 Hour 30 Minutes Marks: 30

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Answer all of the following questions.

1. a) Derive the polynomial equation for 2nd degree with error $\epsilon=0$ for the given dataset [6]
as shown in Table 1. [Polynomial regression is defined as, $y=a_0+a_1x+a_2x^2+\dots+a_nx^n+\epsilon$]

Table 1

Temperature °F	Feeling index (out of 10)
60	9.50
52	8.85
35	6.25
37	7.00
27	5.95

2. a) A researcher has developed a multi-class classification model with four classes: A, B, C, and D. After validation, the following confusion matrix was obtained: [8]

Classes	Actual classes				
Predicted classes	----	A	B	C	D
	A	50	10	5	3
	B	9	61	12	9
	C	5	15	55	52
	D	7	17	57	55

Explain how you can draw the ‘receiver operating characteristic’ curve for a multi-class classification problem.

3. a) We have a sales Cube with 3 dimensions as follows: [8]
 Product $P=\{\text{Laptop, Phone}\}$; Region $R=\{\text{Asia, Europe}\}$; Time $T=\{\text{Q1, Q2}\}$
 We define the sales cube as a function:

$$\text{Sales: } P \times R \times T \rightarrow \mathbb{R}$$

The recorded values are as follows:

Sales(Laptop, Asia, Q1)=5020; Sales(Laptop, Europe, Q1)=6030

Sales(Phone, Asia, Q1)=3070; Sales(Phone, Europe, Q1)=4080

Sales(Laptop, Asia, Q2)=5510; Sales(Laptop, Europe, Q2)=6290

Sales(Phone, Asia, Q2)=3560; Sales(Phone, Europe, Q2)=4240

Using this Cube, answer the following:

- I. ROLL-UP: Compute the total sales per product and time across all regions.
 - II. SLICE: Perform a slice operation to extract the sales cube for **Q1** only.
 - III. DICE: Perform a dice operation to select the sub-cube for Product = {Laptop} and Region = {Asia} across all quarters.
4. a) Dataset in CSV format (Salary_Data.csv) and the code segment are given [2]
 below:

Salary_Data.csv

Years Experience	Salary
1.1	39343
1.3	46205
1.5	37731
2	43525
2.2	39891
2.9	56642
3	60150
3.2	54445

Code segment

```
import numpy as np
import pandas as pd
import seaborn as sns
from sklearn import linear_model
df = pd.read_csv('Salary_Data.csv')
sns.lmplot(x='YearsExperience', y='Salary', data=df)
```

Show the output generated from the code segment.

b) Consider the following dataset, where the last column represents the class label [6]
(evaluation):

maint	doors	size	safety	evaluation
vhigh	2	small	low	unacc
low	3	small	high	acc
vhigh	3	medium	average	unacc
low	3	medium	high	acc
low	4	big	low	unacc
high	4	big	average	acc
low	4	small	average	unacc
low	4	small	high	acc
low	4	medium	low	unacc
low	4	medium	high	acc
low	4	big	low	unacc

Using this dataset, construct a classification tree as a predictive model.